

Data Mining

Introduction

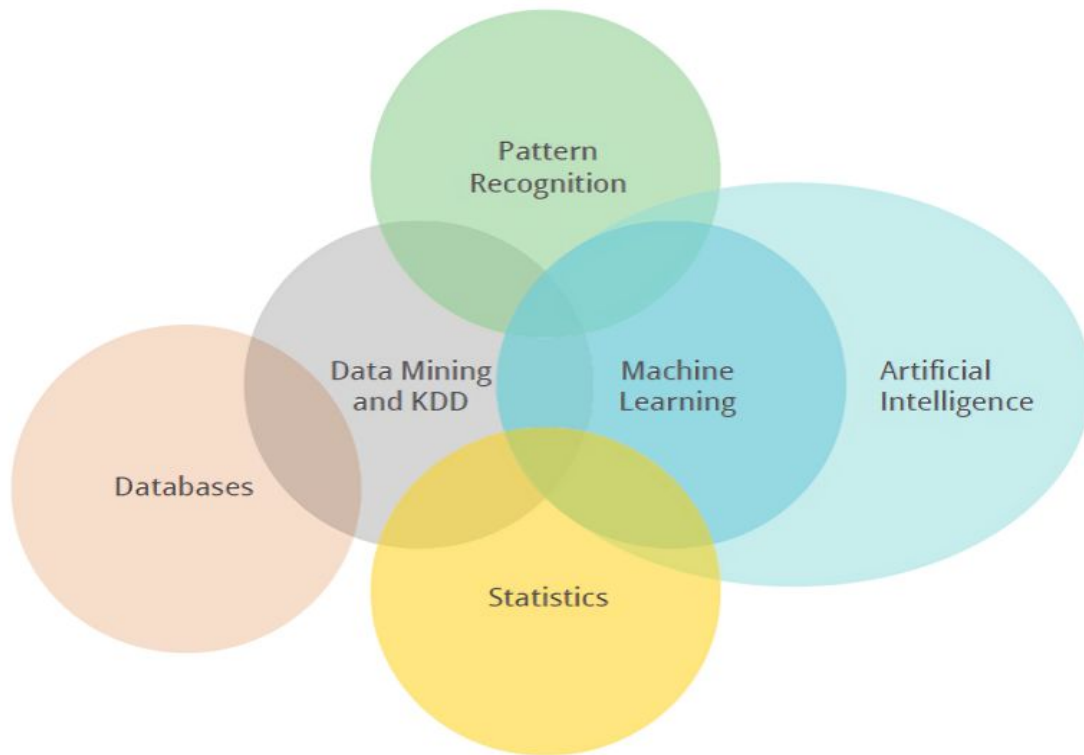
Mohammed Brahimi & Sami Belkacem

Outline

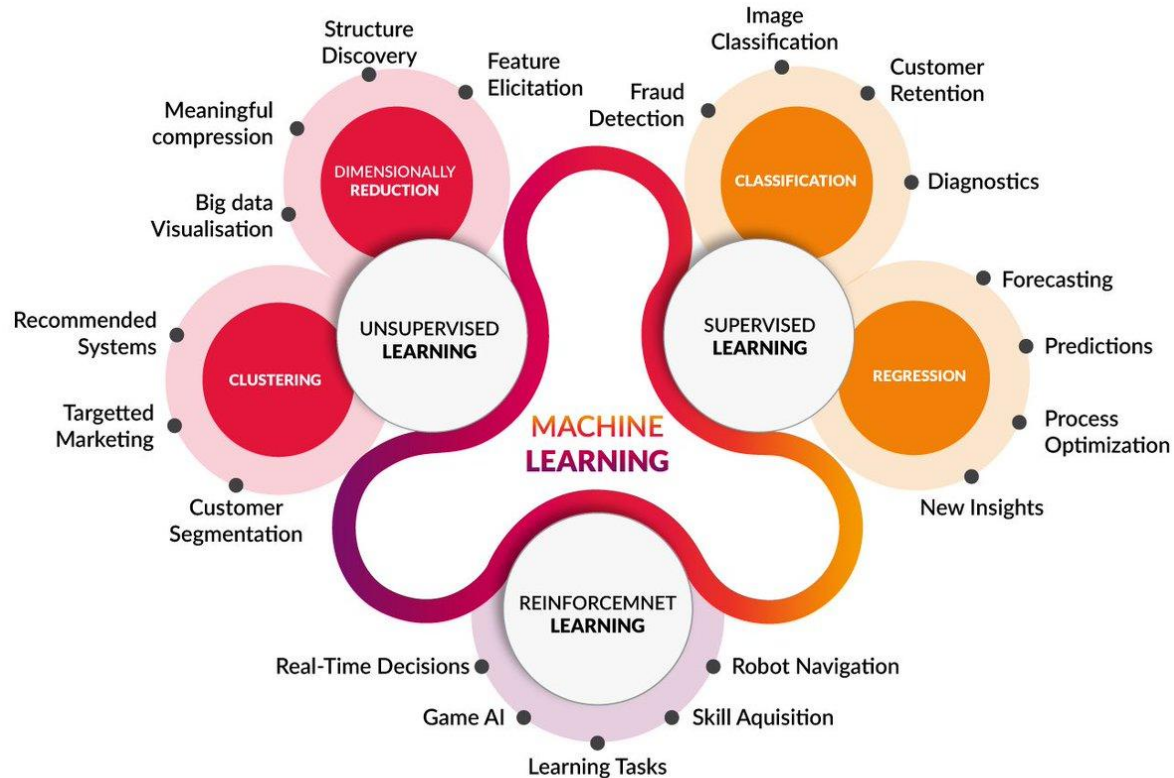
1. Data Mining vs ML vs AI
2. Why Data Mining?
3. What is Data Mining?
4. Data Mining Tasks
5. Motivating Challenges

1. Data Mining vs ML vs AI

Data Mining vs Machine Learning vs AI



Machine Learning



Data Mining vs Machine Learning

	Data Mining	Machine Learning
Goal	Discover Patterns in data to improve decision making	Build and use algorithms that allow computer to learn how to perform certain tasks
Data	Focus mostly on the analysis of current data	Focus on the generalization to unseen data
Insight	Give insights to humans who then decide on business actions	Machines adapts on themselves and learn how to perform tasks on their own

2. Why Data Mining?

Large-Scale Data is Everywhere!

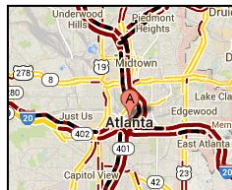
- Data growth in commercial and scientific datasets due to advances in data generation and collection technologies.
- **New mantra:**
 - Gather whatever data you can whenever and wherever possible.
- **Expectations:**
 - Gathered data will have value either for the initial purpose or other purpose not envisioned.



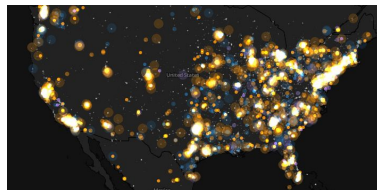
Cyber Security



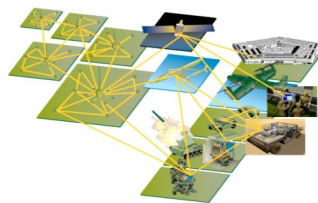
E-Commerce



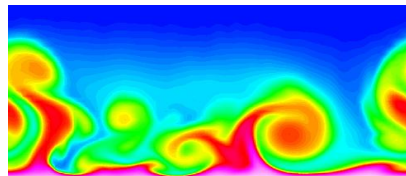
Traffic Patterns



Social Networking: Twitter



Sensor Networks



Computational Simulations

Why Data Mining? Commercial Viewpoint

- Lots of data is collected and warehoused.
 - **Web Data:** Google stores Petabytes of web data.
 - **Social Media:** Facebook has billions of users.
 - **E-commerce:** Millions of daily transactions.
 - **Technological Advancements:** Cheaper and more powerful computers.
- Competitive pressure is strong.
 - **Intense Pressure:** Stiff competition in data era.
 - **Strategic Advantage:** Offer better, customized services (e.g., Customer Relationship Management).

The Facebook logo, consisting of the word "facebook" in white lowercase letters on a blue rectangular background.The Amazon.com logo, featuring the word "amazon.com" in black lowercase letters with a yellow curved arrow underneath.The Yahoo! logo, featuring the word "YAHOO!" in red uppercase letters.The Google logo, featuring the word "Google" in its multi-colored font.

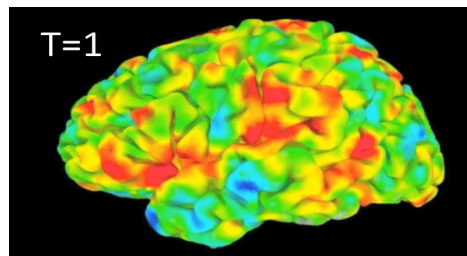
Why Data Mining? Scientific Viewpoint

- Rapid Data Accumulation, e.g. spatial data collection:
 - NASA archives petabytes of earth science data annually.
 - Telescopes scanning the skies.
 - Sky survey data with satellites.

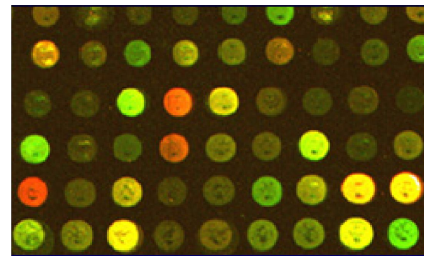


Sky Survey Data

- Biological Insights
 - High-throughput biological data.
- Simulating the Unseen
 - Scientific simulations generate terabytes



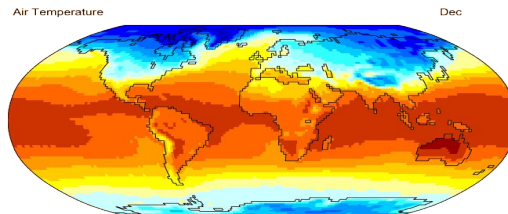
fMRI Data from Brain



Gene Expression Data

Data Mining Empowers Scientists

Automating analysis of massive datasets and aiding hypothesis formation.

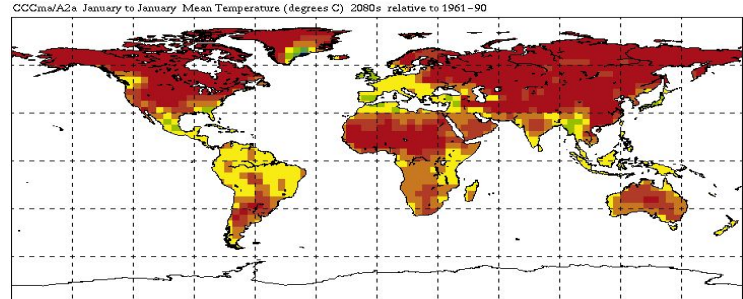


Surface Temperature of Earth

Great Opportunities to Solve Society's Major Problems



Improving healthcare and reducing costs



Predicting the impact of climate change



Finding alternative/green energy sources



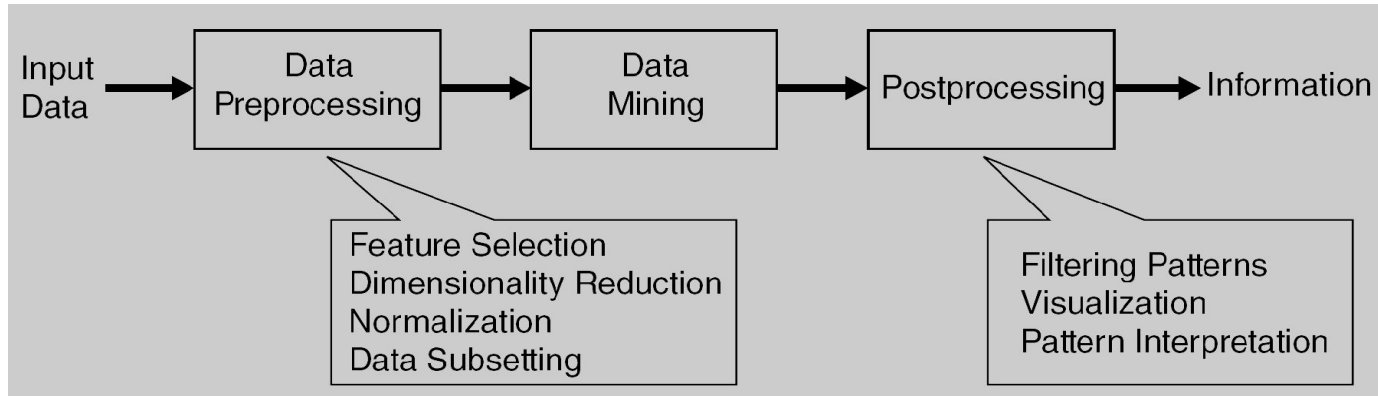
Reducing hunger and poverty by increasing agriculture production

3. What is Data Mining?

Definition of Data Mining

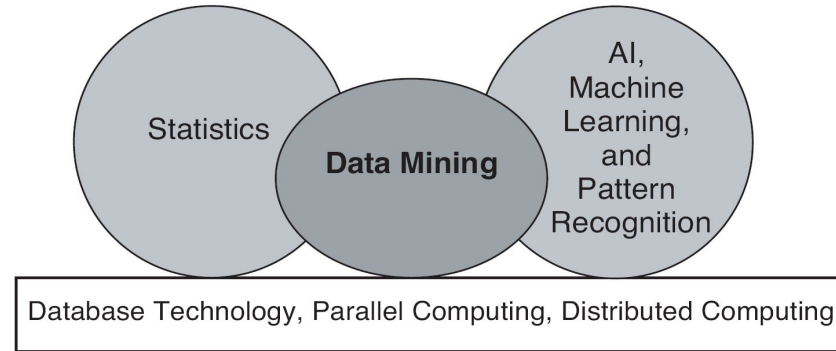
- Many Definitions

- **Extract non-trivial** of implicit, **previously unknown** and **potentially useful** information from data.
- **Exploration and analysis**, by automatic or semi-automatic means, of **large quantities of data** in order to **discover meaningful patterns**.



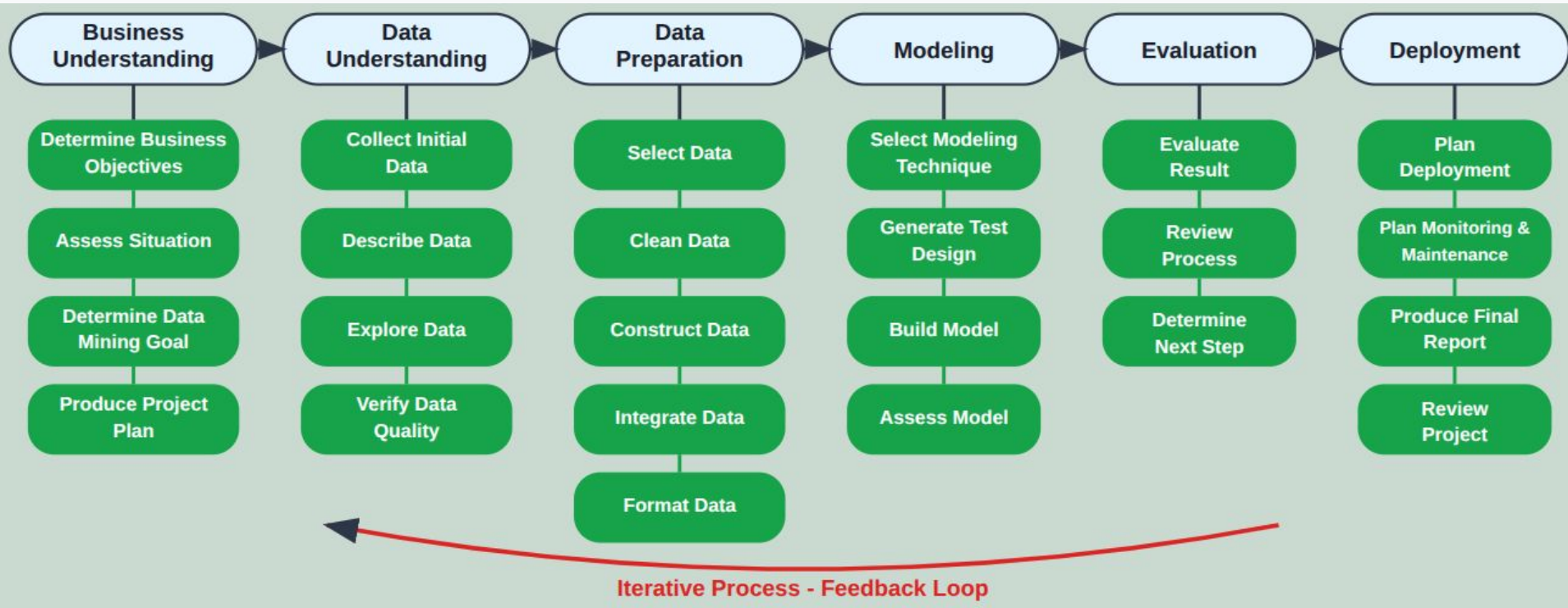
Origins of Data Mining

- Ideas from many fields
 - machine learning/AI, pattern recognition, statistics, and database systems
- Traditional techniques are unsuitable due to the characteristics of the data:
 - Large-scale, High dimensional
 - Heterogeneous, Complex
 - Distributed



A key component of the emerging field of data science and data-driven discovery

Implementation Process of Data Mining



4. Data Mining Tasks

Data Mining Tasks

- **Prediction Methods**

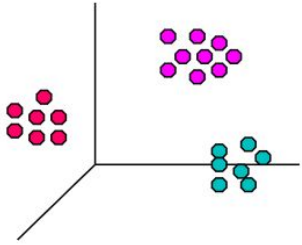
- Use some variables to predict unknown or future values of other variables.
- **Example:** Sales forecasting in E-commerce



- **Description Methods**

- Find human-interpretable patterns that describe the data.
- **Example:** Analyzing historical criminals data for profiling.

Data Mining Tasks

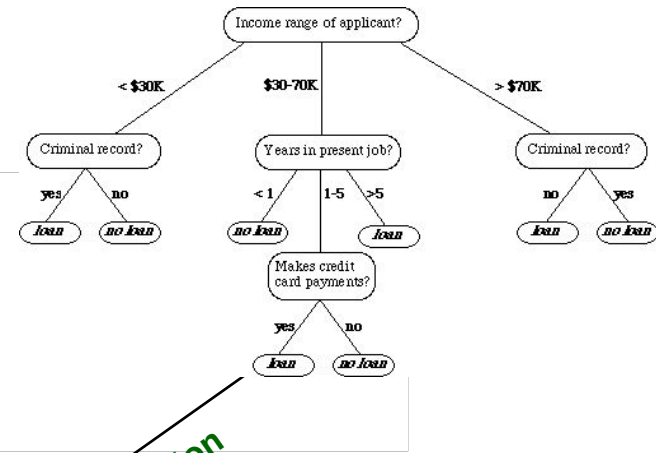


Clustering

Association Rules

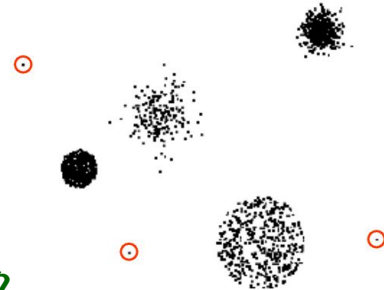
{Refund, Married} → Cheat

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes
11	No	Married	60K	No
12	Yes	Divorced	220K	No
13	No	Single	85K	Yes
14	No	Married	75K	No
15	No	Single	90K	Yes



Classification

Anomaly Detection



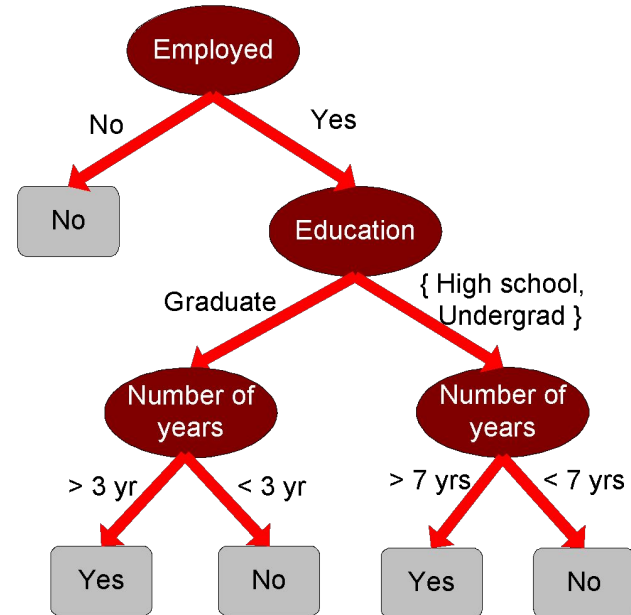
Classification

- Find a model for class attribute as a function of the values of attributes.

Model for predicting credit worthiness

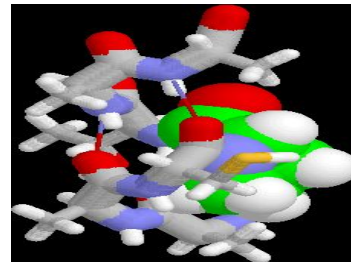
Class

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...



Examples of Classification Tasks

- Credit card transaction classification
 - Legitimate vs. fraudulent.
- Satellite data
 - Land cover classification (water bodies, urban areas, . etc.).
- News story categorization
 - Finance, weather, entertainment, sports, etc.
- Cyberspace security
 - Intruder identification.
- Medical diagnosis
 - Tumor cell classification (benign vs. malignant).
- Protein analysis
 - Classification of secondary structures (alpha-helix, beta-sheet, or random coil).



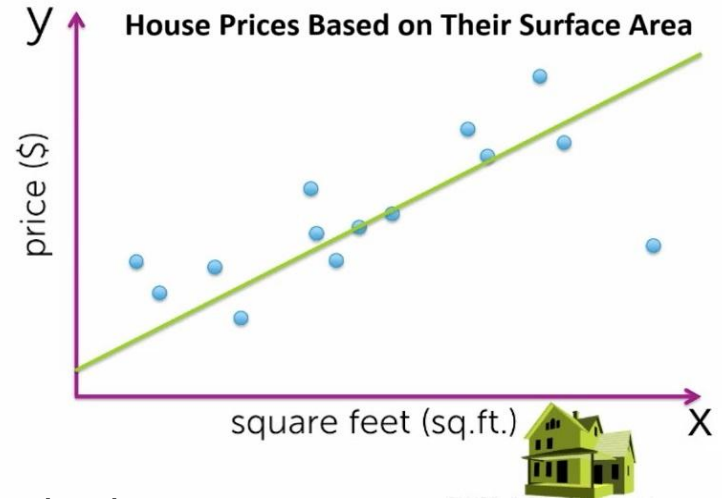
Applications of Classification

Imagine a use case and recommend a classification-based approach to resolve it.



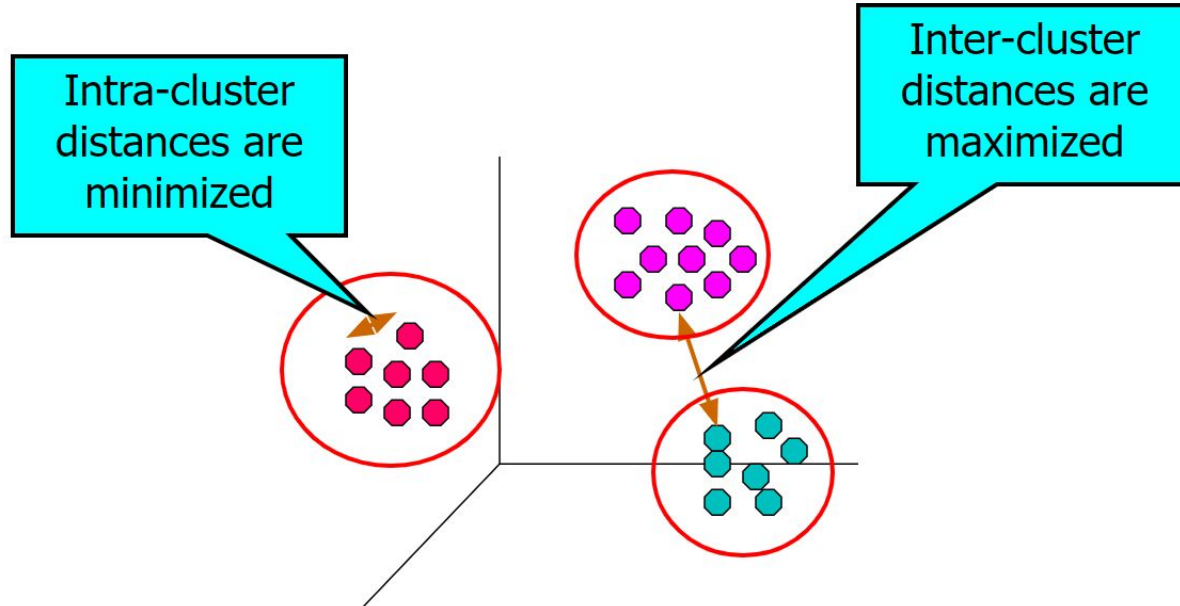
Regression

- **Predicting a continuous variable by considering the relationships with other variables, using linear or nonlinear models.**
- **Examples:**
 - **Forecasting Sales:** Predicting new product sales by analyzing advertising expenditure.
 - **Wind Velocity Prediction:** Estimating wind velocities using variables like temperature, humidity, and air pressure.
 - **Stock Market Forecast:** Predicting stock market indices through time series analysis.



Clustering

Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



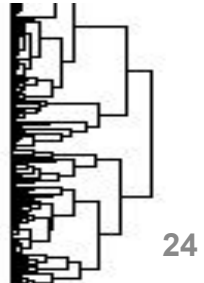
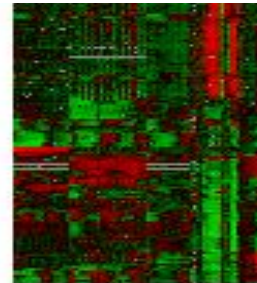
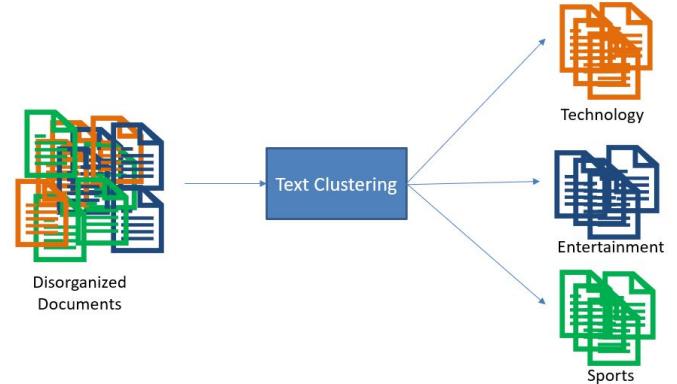
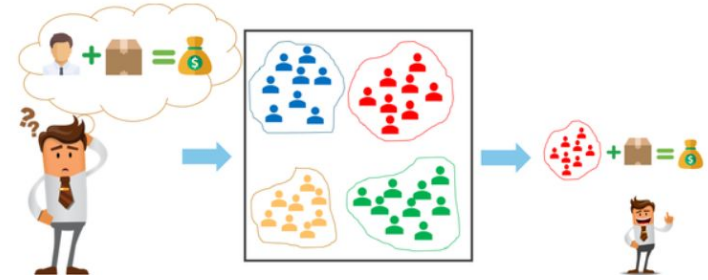
Applications of Clustering

- **Understanding**

- Custom profiling for targeted marketing
- Group related documents for browsing
- Group genes that have similar functionality
- Group stocks with similar price fluctuations

- **Summarization**

- Reduce the size of large data sets



Association Rules

- Given a set of records each of which contain some number of items from a given collection.
 - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Juice, Coke, Milk
2	Juice, Bread
3	Juice, Coke, Diaper, Milk
4	Juice, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

{Milk} --> {Coke}

{Diaper, Milk} --> {Juice}

Applications of Association Rules

- **Market-basket analysis**

- Rules are used for sales promotion, shelf management, and inventory management

- **Telecommunication alarm diagnosis**

- Rules are used to find combination of alarms that occur together frequently in the same time period

- **Medical Informatics**

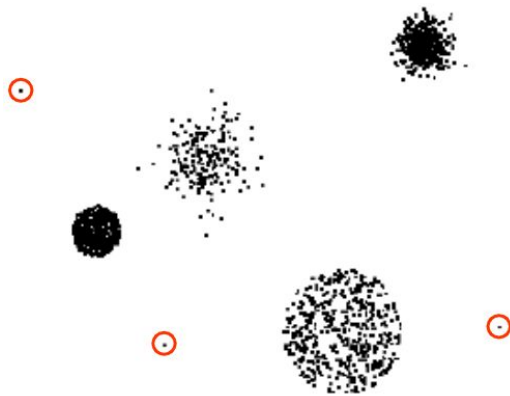
- Rules are used to find combination of patient symptoms and test results associated with certain diseases

Deviation/Anomaly/Change Detection

Detect significant deviations from normal behavior

- **Applications:**

- Detecting changes in the global forest cover
- Credit Card Fraud Detection
- Network Intrusion Detection
- Identify anomalous behavior from sensor networks for monitoring and surveillance.



5. Motivating Challenges

Motivating Challenges

- **Scalability:**
 - Efficiently managing and processing massive datasets.
- **High Dimensionality:**
 - Addressing complex data structures with numerous attributes.
- **Heterogeneous and Complex Data:**
 - Handling diverse data types and intricate data structures.
- **Data Ownership and Distribution:**
 - Tackling geographically distributed data owned by multiple entities.
 - Challenges: minimize communication, consolidate results, and ensure data security and privacy.
- **Non-traditional Analysis:**
 - Statistical methodology is unable to deal with current data nature.

Understanding Question (1)

- These Data Mining techniques are Descriptive or Predictive, and why?
 - Classification
 - Clustering
 - Association Rules
 - Anomaly detection

Understanding Question (2)

- Which Data Mining technique is suitable, and why?
 - Detect defective products in a production chain of a factory
 - Predict which customers are likely to cancel their subscriptions
 - Analyzes viewing habits in a streaming platform to find patterns
 - Group similar products together based on some features
 - Categorize emails into predefined categories (Spam/not Spam)
 - Discover patterns in user online navigation without predefined groups
 - Analyzes patient vital signs to detect sudden changes